

SQL Join Types

In SQL, joins are used to combine rows from two or more tables based on a related column between them. Here are the main types of joins:

1. INNER JOIN: Returns records that have matching values in both tables.

```
```sql
SELECT columns
FROM table1
INNER JOIN table2
ON table1.column = table2.column;
```
```

2. LEFT (OUTER) JOIN: Returns all records from the left table (table1), and the matched records from the right table (table2). If there is no match, the result is NULL on the side of the right table.

```
```sql
SELECT columns
FROM table1
LEFT JOIN table2
ON table1.column = table2.column;
```
```

3. RIGHT (OUTER) JOIN: Returns all records from the right table (table2), and the matched records from the left table (table1). If there is no match, the result is NULL on the side of the left table.

```
```sql
SELECT columns
FROM table1
RIGHT JOIN table2
ON table1.column = table2.column;
```
```

4. FULL (OUTER) JOIN: Returns all records when there is a match in either left (table1) or right (table2) table. Records that do not match are also included, with NULLs in place of missing data.

```
```sql
SELECT columns
FROM table1
FULL OUTER JOIN table2
ON table1.column = table2.column;
```
```

5. CROSS JOIN: Returns the Cartesian product of the two tables, meaning it combines each row of the first table with each row of the second table.

```
```sql
SELECT columns
FROM table1
CROSS JOIN table2;
```
```

6. SELF JOIN: A regular join, but the table is joined with itself. It's useful for querying hierarchical data or comparing rows within the same table.

```
```sql  

SELECT a.columns, b.columns

FROM table1 a, table1 b

WHERE condition;
```
```

7. NATURAL JOIN: A type of join that automatically joins tables based on columns with the same names and compatible types. It can be an INNER JOIN, LEFT JOIN, or RIGHT JOIN.

```
```sql  

SELECT columns

FROM table1

NATURAL JOIN table2;

WHERE condition;(Optional)
```
```

Each type of join serves a different purpose and can be used depending on the specific requirements of the data retrieval operation.

View

A **view** in SQL is a **virtual table** that is defined by a SQL query. Unlike a real table, a view does not store data physically. Instead, it provides a dynamic result set based on the query it encapsulates.

```
```Sql
```

```
CREATE VIEW view_name AS
```

```
SELECT column1, column2, ...
```

```
FROM table_name
```

```
WHERE condition;
```

```
```
```

- **view_name:** The name of the view you want to create.
- **SELECT statement:** Defines the columns and rows included in the view. It can include filters, joins, aggregations, and other SQL features.

For Updating view

```
```Sql
```

```
UPDATE view_name
```

```
SET column1 = value1, column2 = value2, ...
```

```
WHERE condition;
```

```
```
```